

Exploratory Data Analysis (EDA) & Business Intelligence Report

Prepared By

Manogna Akula

Internship

ApexPlanet Data Analytics Internship

Task 2: Exploratory Data Analysis (EDA) & Business Intelligence

1. Introduction

Exploratory Data Analysis (EDA) is an important step in the data analytics process that helps identify patterns, trends, and relationships within a dataset. The objective of this project is to analyze sales transaction data and generate meaningful business insights through statistical analysis and visualizations.

The dataset contains information related to customer orders, product categories, sales revenue, geographical locations, and deal sizes. Various analytical techniques were applied to understand customer behavior, product performance, and sales trends.

2. Dataset Overview

The dataset consists of sales transaction records containing customer, product, and order-related information.

Key Attributes

- Order Number
- Quantity Ordered
- Price Each
- Sales
- Order Date
- Product Line
- Customer Name
- Country
- Deal Size

Dataset Size

- Total Records: 2823
- Total Columns: 30

The dataset was cleaned and prepared during Task-1 before performing exploratory analysis.

3. Descriptive Statistics

Descriptive statistics were calculated to understand the overall distribution of numerical variables.

Key Findings

- Average Sales Value: 3553.89
- Minimum Sales Value: 482.13
- Maximum Sales Value: 14082.80
- Average Quantity Ordered: 35 Units
- Average Product Price: 83.66

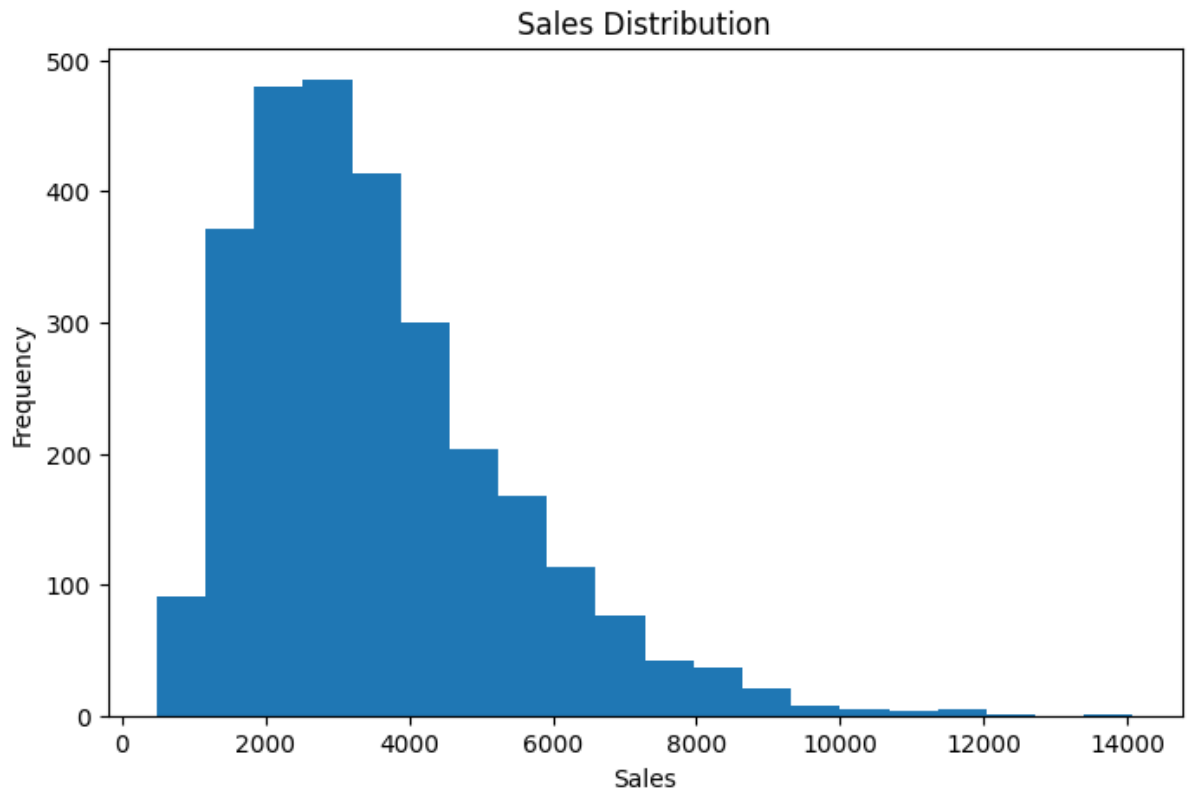
The statistics indicate significant variation in transaction values, with both low-value and high-value sales present in the dataset.

4. Sales Distribution Analysis

A histogram was created to analyze the distribution of sales transactions.

Observations

- Most sales transactions fall between 1500 and 4500.
- High-value transactions are less frequent.
- A small number of orders generated sales above 8000.
- The distribution is positively skewed (right-skewed).



Business Insight

A small percentage of transactions contribute significantly to total revenue, indicating the presence of premium customers and bulk purchases.

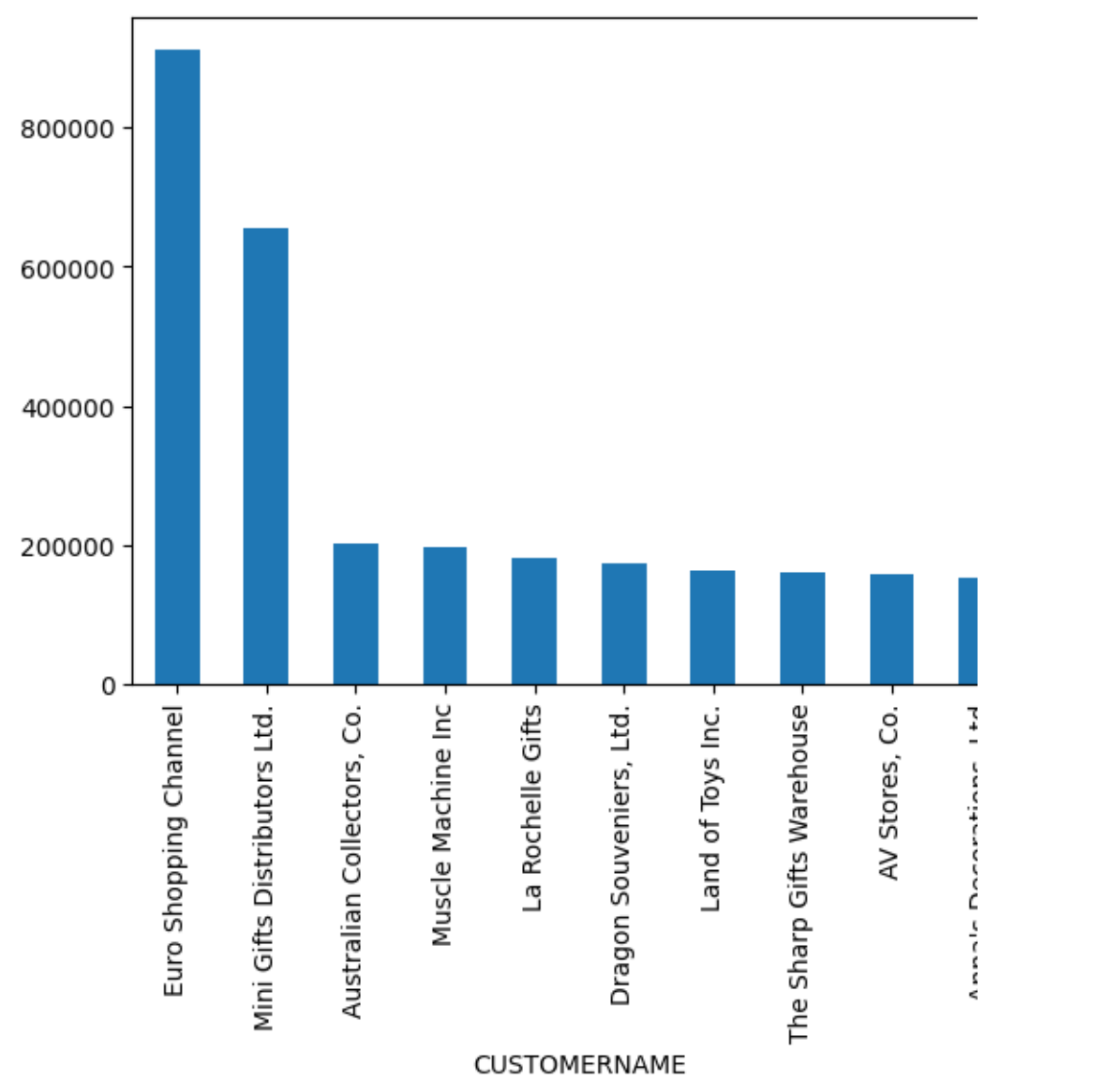
5. Customer Analysis

Customer-wise sales analysis was performed to identify the highest revenue-generating customers.

Top 5 Customers

Customer Name	Sales
Euro Shopping Channel	912,294.1 1
Mini Gifts Distributors Ltd.	654,858.0 6
Australian Collectors, Co.	200,995.4 1

Customer Name	Sales
Muscle Machine Inc	197,736.94
La Rochelle Gifts	180,124.90



Business Insight

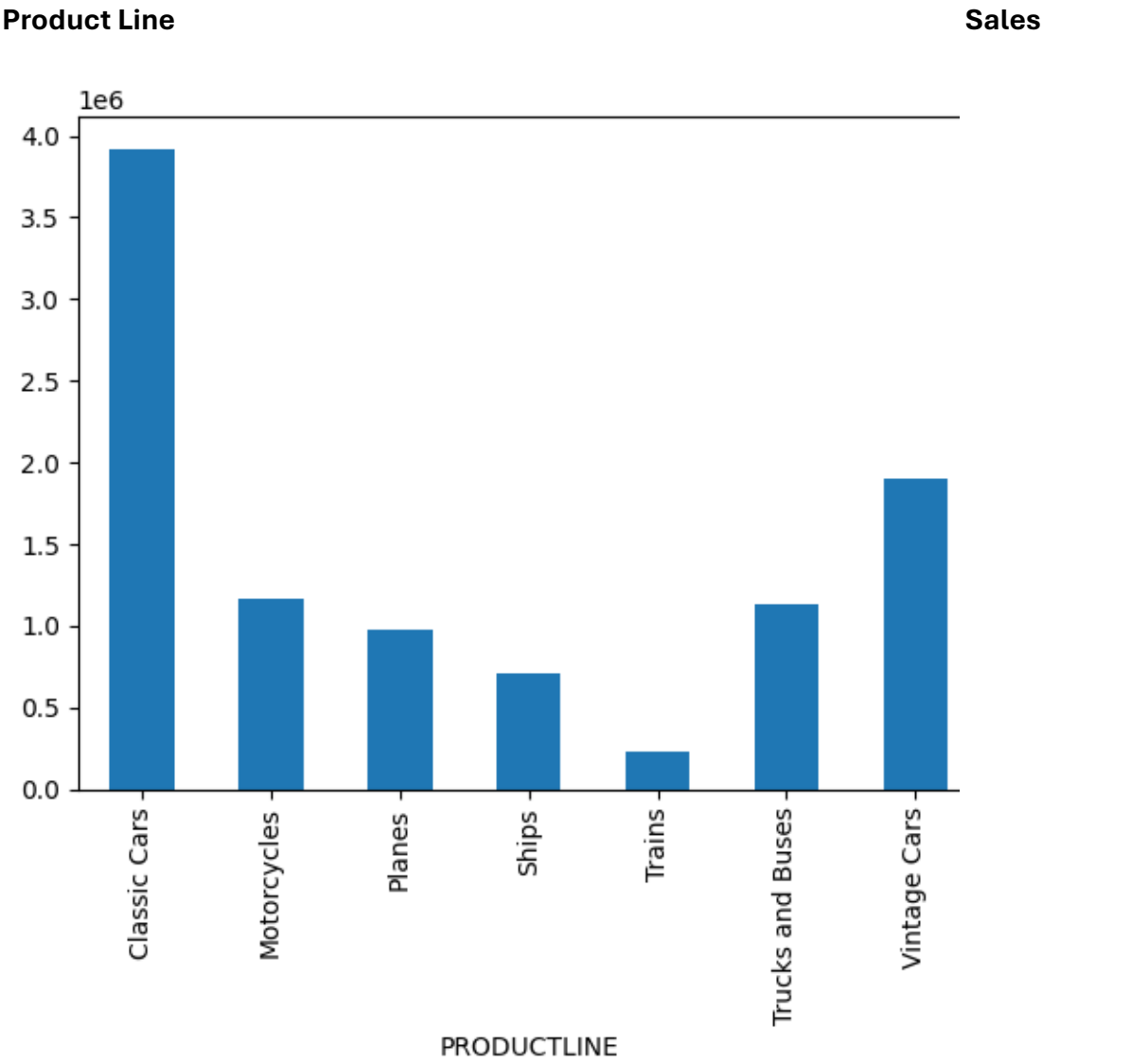
Euro Shopping Channel is the highest revenue-generating customer. The top customers contribute a substantial share of total revenue, highlighting the importance of customer retention strategies.

6. Product Line Analysis

Sales performance was analyzed across different product categories.

Product Line Revenue

Product Line	Sales
Classic Cars	3,919,615.66
Vintage Cars	1,903,150.84
Motorcycles	1,166,388.34
Trucks and Buses	1,127,789.84
Planes	975,003.57
Ships	714,437.13
Trains	226,243.47



Business Insight

Classic Cars generated the highest revenue among all product categories. This category serves as the primary revenue driver for the business.

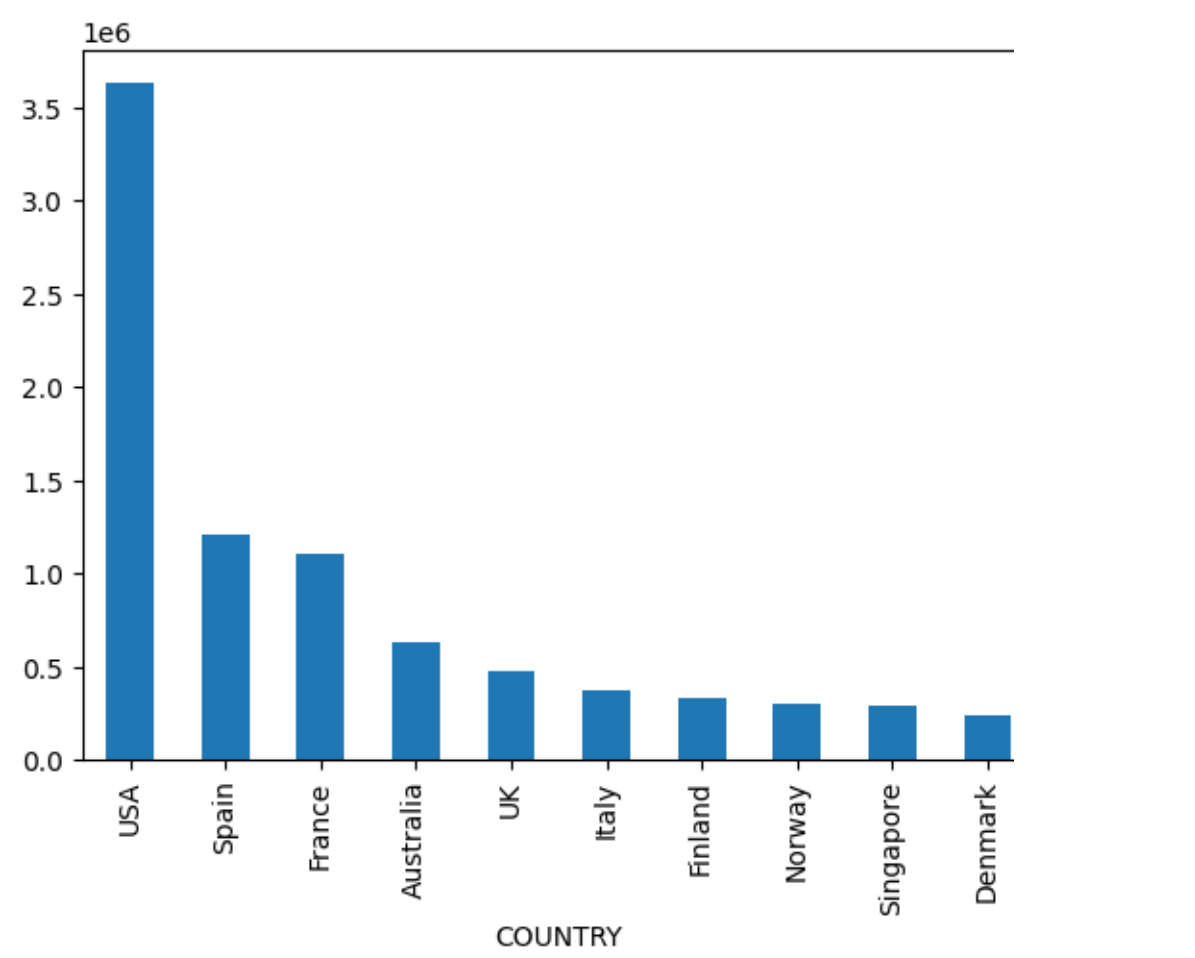
7. Country-wise Sales Analysis

Sales revenue was analyzed across different countries.

Top Countries by Sales

Country	Sales
USA	3,627,982.83

Country	Sales
Spain	1,215,686.92
France	1,110,916.52
Australia	630,623.10
UK	478,880.46



Business Insight

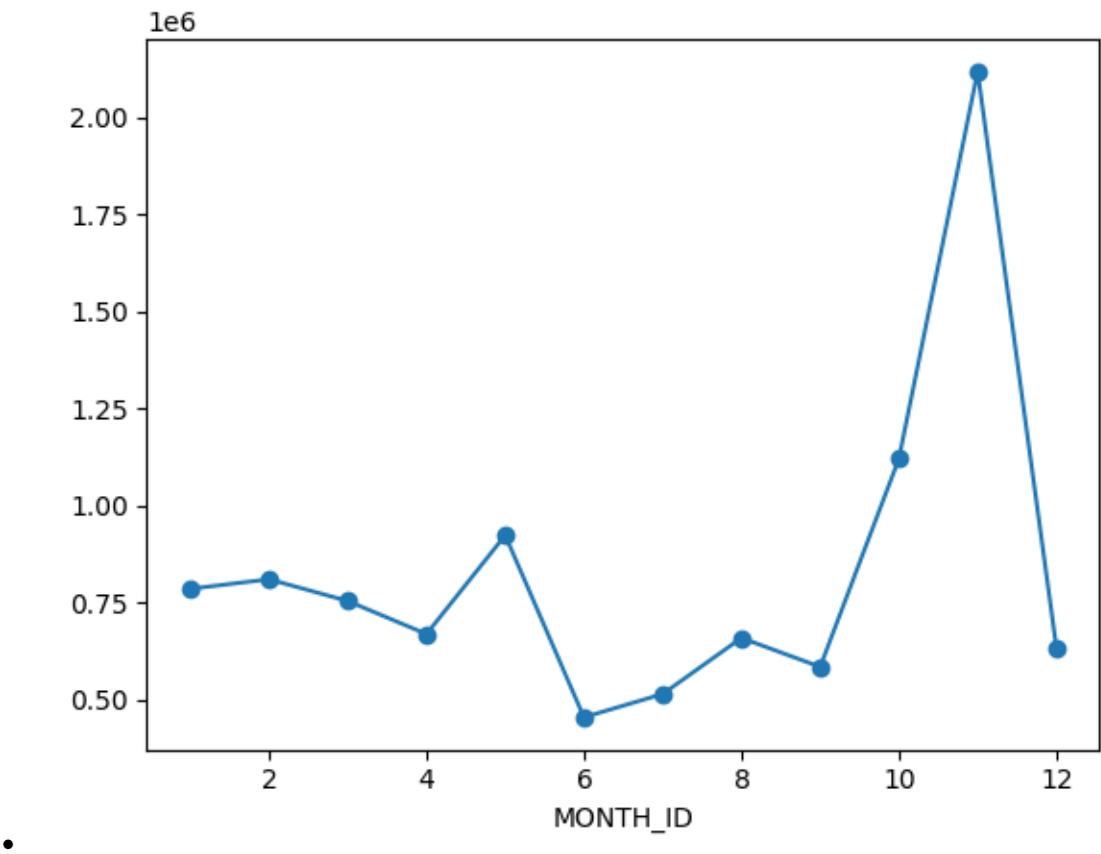
The USA is the highest revenue-generating market. International markets such as Spain and France also contribute significantly to total sales revenue.

8. Monthly Sales Trend Analysis

Monthly sales performance was analyzed to identify seasonal patterns.

Key Findings

- Highest Sales Month: November
- Revenue in November: 2,118,885.67
- Lowest Sales Month: June
- Revenue in June: 454,756.78



Business Insight

Sales increased significantly during the last quarter of the year, suggesting seasonal demand, promotional campaigns, or holiday-related purchasing behavior.

9. Deal Size Analysis

Revenue contribution by deal size was examined.

Revenue by Deal Size

Deal Size	Sales
Medium	6,087,432.24

Deal Size

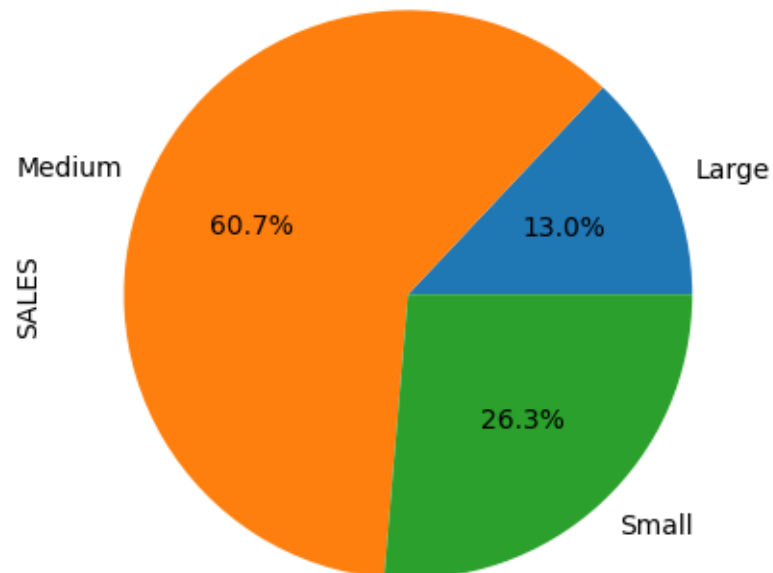
Small

Large

Sales

2,643,077.35

1,302,119.26



Business Insight

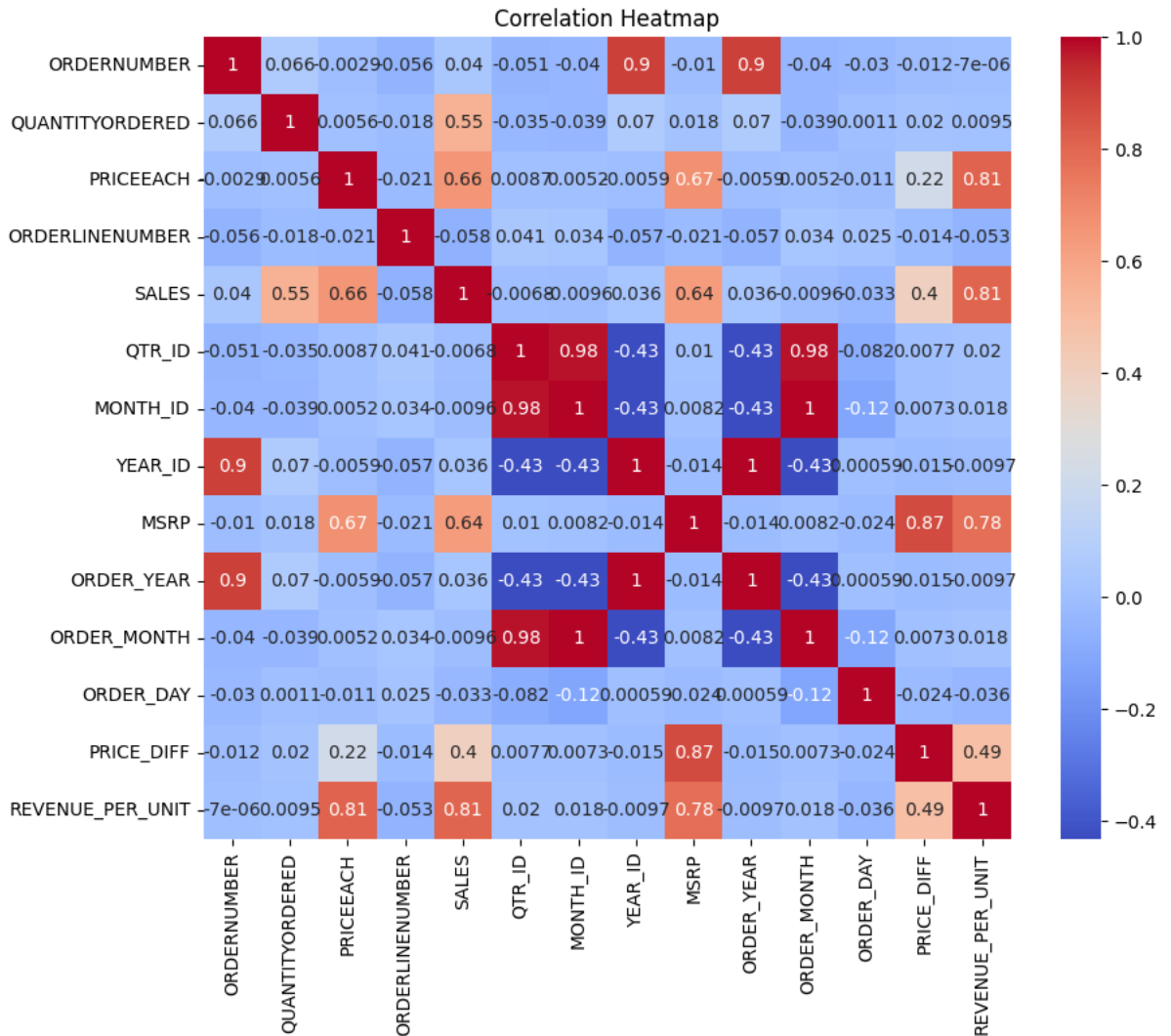
Medium-sized deals generate the majority of revenue and represent the most important transaction category for the business.

10. Correlation Analysis

A correlation heatmap was generated to examine relationships between numerical variables.

Key Findings

- SALES and REVENUE_PER_UNIT show strong positive correlation (0.81).
- QUANTITYORDERED and SALES show moderate positive correlation (0.55).
- MSRP and PRICEEACH are positively correlated.



Business Insight

Sales performance is strongly influenced by revenue per unit and order quantity, indicating that pricing and volume significantly impact overall revenue.

11. Business Questions and Answers

Q1. Who is the highest revenue-generating customer?

Euro Shopping Channel generated the highest revenue with sales of 912,294.11.

Q2. Which product line generates the highest revenue?

Classic Cars generated the highest revenue with sales exceeding 3.91 million.

Q3. Which country contributes the highest revenue?

USA contributes the highest sales revenue with approximately 3.63 million in sales.

Q4. Which month recorded the highest sales?

November recorded the highest sales with revenue of 2.11 million.

Q5. Which deal size contributes the most revenue?

Medium-sized deals contribute the highest revenue with total sales of 6.09 million.

12. Conclusion

The Exploratory Data Analysis successfully identified key revenue drivers, customer trends, and product performance patterns within the sales dataset.

The analysis revealed that Classic Cars is the most profitable product category, USA is the strongest market, and Euro Shopping Channel is the most valuable customer.

Seasonal sales trends indicate strong business performance during the final quarter of the year, while medium-sized deals contribute the majority of total revenue.

These insights can help management improve customer retention, optimize product strategies, and support data-driven business decision-making.